# Age-Sensitive Privacy Expectations on Social Media: A Contextual Integrity Analysis

Yuxing Yang
School of Information Sciences, University of Illinois Urbana-Champaign
Champaign, IL, USA
e-mail: yuxingy2@illinois.edu

***Abstract*—Social media platforms encode privacy through visibility settings, consent tools, recommendation systems, and data sharing controls, and they usually offer every user the same choices. Privacy expectations, however, are not uniform. Teens, young adults, and older users often read audience limits, content storage, and later data use in different ways, even on the same platform. This work in progress uses Contextual Integrity to study that gap, treating age not only as a legal label but as a way to understand how users interpret information flows. We call the underlying pattern normative flattening: a platform applies one privacy baseline, usually that of a digitally skilled adult, to users with different skills, risks, and expectations. We relate this idea to context collapse and networked publics, propose a four-step analytic framework, and apply it to a case on teen visibility and algorithmic amplification. The aim is to move privacy analysis from formal compliance toward how users actually understand and experience platform design.**

***Keywords-social media privacy; age-sensitive design; contextual integrity; context collapse; privacy expectations; platform governance; normative flattening.***

## I. Introduction

Social media platforms are shared spaces. Users at different life stages meet the same interface, the same settings, and the same data sharing rules. A teenager may use a platform to manage peer ties and identity. A young adult may use it for school, work, friendship, and public expression. An older user may use it for family, local groups, and public information. They face the same controls, but they do not always understand them in the same way.

Recent platform and policy changes show why age matters here. Meta introduced Instagram Teen Accounts in 2024 with stronger defaults for younger users [1]. The United Kingdom Age-Appropriate Design Code asks online services to consider children when setting defaults [2]. Australia passed a law setting a minimum age for social media accounts [3]. Each step concedes that one privacy setting does not fit every age group.

Most platforms still frame privacy as a matter of user choice: make an account public or private, accept a policy, decide who sees a post. These controls matter, but they reveal little about how people understand privacy in daily use. A user may accept a policy without knowing how data will be reused, choose an audience without seeing how recommendation widens reach, or post for a brief social moment that the platform then stores, ranks, and redistributes [4][5].

We argue that age shapes how users read these controls. As a legal or demographic category, age supports compliance but explains little about experience. Teens tend to focus on peer visibility and social judgment [6][7]. Young adults weigh future reputation and the line between social and professional life. Older users more often meet scams, false information, and settings they find hard to read. These are tendencies, not fixed traits, since age works together with skill, culture, and past experience [8]. Still, age helps reveal where one design choice carries different meanings for different users.

We study the problem through Contextual Integrity, which defines privacy as appropriate information flow within a social context [9][10]. It asks who sends information, who receives it, what is shared, who it concerns, and under what condition it moves. This view fits social media, where privacy problems often come from a gap between expected and actual flow. We name one such pattern *normative flattening*: a platform applies a single privacy baseline, often that of a digitally skilled adult, to users with different skills, risks, and expectations. We do not add age as a sixth Contextual Integrity parameter. We use it as a lens for finding mismatches in shared social media spaces.

Prior work has examined contextual privacy norms, collapsed audiences, networked publics, and age-related differences in digital skills and platform design. However, these strands do not directly explain how the same platform-mediated information flow may carry different privacy meanings for users at different life stages while being governed by a shared interface and privacy baseline. This gap motivates an age-sensitive analysis of the assumptions embedded in social media privacy design.

Accordingly, this work asks: How can Contextual Integrity be used to identify age-related mismatches between users' expected information flows and platform-mediated information flows on social media? This work in progress makes three contributions:

- **C1 — Conceptual contribution:** It defines normative flattening and distinguishes it from context collapse and networked publics;
- **C2 — Analytical contribution:** It proposes a four-step framework for age-sensitive privacy analysis based on Contextual Integrity; and
- **C3 — Applied contribution:** It applies the framework to teen visibility and algorithmic amplification to demonstrate how it can guide design and policy analysis.

This work is conceptual and does not yet provide empirical user-study evidence across age groups. The case presented in Section VI should therefore be understood as an illustrative application of the framework rather than as empirical validation. The framework requires further testing through scenario-based studies, interviews, and platform audits.

The paper proceeds as follows. Section II reviews Contextual Integrity and age-sensitive privacy. Section III places the

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argument in related work. Section IV defines normative flattening. Section V presents the framework. Section VI applies it to a case. Section VII gives limits and future work.

## II. Background: Contextual Integrity and Age-Sensitive Privacy

Contextual Integrity treats privacy as the appropriate flow of information within a social context, not only as secrecy or personal control [9][10]. It studies five elements: sender, recipient, subject, attribute, and transmission principle. The transmission principle is the condition for movement, such as consent, confidentiality, or service need.

Social media makes these flows hard to see. A user may post a photo, video, or comment to a chosen audience. After that, the platform may store, rank, recommend, and redistribute the content. The user sees the first act of sharing but rarely the longer chain that follows. Consent and visibility settings therefore capture only part of the privacy problem.

We use Contextual Integrity in a practical way. The original theory asks whether a flow fits the shared norms of a context. We also ask how users interpret the same flow. A user's expectation is not the same as a norm, but repeated gaps between interpretation and actual platform behavior point to a likely mismatch, and those gaps show where design may fail users.

We therefore treat user expectations as empirical signals of possible contextual mismatch rather than as contextual norms themselves. Whether such mismatches reflect broader normative violations requires subsequent empirical validation.

Age matters because users at different life stages may read the same flow differently. Teens often use social media for peer ties, identity, and belonging, and may focus on peer visibility and social judgment [6][7]. Young adults move across school, work, friendship, and public life, and may worry about role boundaries and future reputation. Older users connect with family and community, and may meet confusing settings, unwanted exposure, scams, and false information.

We do not claim that everyone in an age group thinks alike. Culture, education, gender, class, disability, and prior experience also matter [8]. The claim is narrower. Age is useful because platforms and laws already rely on it in account design, safety tools, and privacy rules, yet they tend to use broad age bands. We ask whether those bands hide different privacy interpretations. Age therefore functions here as a comparative analytic lens rather than as an additional parameter of the Contextual Integrity framework.

## III. Related Work

This work builds on four areas and clarifies how normative flattening differs from earlier ideas.

*Context collapse and the imagined audience.* Marwick and Boyd show that social media brings distinct audiences into one space, so users address an imagined audience that may not match the real one [7][11]. Context collapse explains how audiences merge within a single act of sharing. Its primary focus, however, is audience convergence rather than the privacy assumptions embedded in platform design. Normative flattening shifts the analysis from who is present in the audience to whose understanding of information flow is implicitly assumed by the platform. In this sense, it complements context collapse by identifying a distinct source of mismatch: a shared privacy baseline applied to users who may interpret the same flow differently.

*Networked publics.* Boyd describes networked publics through persistence, visibility, spreadability, and searchability [6][12]. These properties explain how content behaves once posted and why social media privacy is hard. Their primary strength is explaining how platform affordances can extend the reach and persistence of information beyond the user's immediate social setting. They do not, however, directly examine whether users at different life stages interpret those affordances in the same way. Normative flattening builds on this account by focusing on interpretive differences: the same platform affordance may carry different privacy meanings for users with different expectations, skills, and social contexts.

*Contextual Integrity as an analytic and empirical framework.* Contextual Integrity provides a structured way to represent privacy through the sender, recipient, subject, attribute, and transmission principle of an information flow [13][14]. Prior work has shown that these contextual parameters can be formalized and empirically examined rather than used only as a conceptual description of privacy. Our framework retains this flow-based structure but adds a comparative question: whether users at different life stages interpret the same platform-mediated flow in similar ways. Age is therefore used to identify possible differences in interpretation without altering the underlying Contextual Integrity parameters.

*Age-appropriate design and digital skill.* Recent rules and platform changes already treat age as relevant to privacy design. Age-appropriate design codes ask services to set safer defaults for children [2], Instagram Teen Accounts apply stronger defaults for younger users [1], and access laws use age as a policy line [3]. Work on digital skill shows that access to a tool does not grant equal ability to use it safely [8].

These approaches establish that age and digital competence can matter for privacy protection, but they largely address regulation, access, or protective design interventions. They provide less guidance for examining how users at different life stages may assign different privacy meanings to the same information flow. Normative flattening connects these concerns to platform-level privacy analysis by asking whether a shared interface and privacy baseline implicitly assume a level of understanding that is not shared equally across users.

Taken together, prior research explains contextual privacy norms, audience convergence, platform affordances, digital skill, and age-based protective interventions. What remains less developed is an analytic account of how a single platform-mediated information flow may carry different privacy meanings across age groups while being governed by a shared interface and privacy baseline. Normative flattening addresses this gap by connecting differences in user interpretation to the assumptions embedded in platform privacy design.

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## IV. Normative Flattening in Social Media Privacy

Social media platforms often give the same privacy tools to a wide range of users. The same account options, consent dialogs, and sharing controls appear for everyone, which can look fair. But equal settings do not always produce equal privacy, because users bring different skills, risks, and expectations to the same tool.

Normative flattening occurs when a platform treats different users as if they shared one set of privacy expectations. It applies a single baseline for visibility, audience, persistence, and data reuse. That baseline can hide age related differences, and it can make privacy harms look like user mistakes even when the design helped create the mismatch. The concept does not assume that all users within an age group share the same expectations, nor does it imply that every uniform privacy setting is necessarily harmful. Rather, it identifies cases in which a shared design baseline may obscure meaningful differences in how users understand an information flow. We describe four common forms.

*Visibility flattening.* Platforms use broad labels such as public, private, friends, or followers. The labels look simple but read differently across users. A teen may treat followers as a known peer circle, while an older user may not see how tagging, group settings, or profile settings widen visibility.

*Consent flattening.* Platforms often treat one click as agreement. Clicking agree or continuing to use a service need not mean the user understands storage, ranking, recommendation, reuse, or third-party sharing [4][5].

*Audience flattening.* Users post to an imagined audience of friends, classmates, family, or a local group. Recommendation, search, screenshots, tagging, and resharing can carry content outside that group. The harm appears when the actual audience is far wider than the expected one.

*Persistence flattening.* Users may share for a brief moment, such as a joke or personal update meant for one setting. The platform can keep, rank, and resurface that content, and others can save or repost it. A short moment can become lasting exposure.

Table I links the four forms to platform features and the age-based mismatch each can produce. Together they show why formal compliance is not enough: a platform can follow its stated rules and still create mismatches. What matters is not only whether users have choices, but whether those choices match how users understand the flow.

As an analytic concept, normative flattening has two practical advantages. It is compatible with the existing Contextual Integrity parameters, and it links abstract privacy expectations to concrete platform features that can be examined through design reviews, scenario studies, or platform audits. Its limitations are equally important: the framework does not by itself establish causal differences between age groups, and age is only one source of variation in privacy expectations. Empirical validation is therefore needed to determine when the identified mismatches are systematic rather than case-specific.

TABLE I. Forms of Normative Flattening

| Form | Related feature | Possible age-based mismatch |
|---|---|---|
| Visibility | Visibility | Simple labels can mean a known circle to some users and broad reach to others |
| Consent | Consent tools | One click can be treated as informed agreement to later use |
| Audience | Sharing and search | A peer audience can become a wider audience through recommendation and resharing |
| Persistence | Storage | A short purpose can become long-term visibility |

TABLE II. Four-Step Age-Sensitive Analysis

| Step | Main question | Purpose |
|---|---|---|
| 1 | Who sends what information to whom, and under what condition? | Separate intended sharing from later movement |
| 2 | What user does the design assume? | Reveal the baseline in the design |
| 3 | Do age groups read the flow differently? | Find possible age-based mismatches |
| 4 | What design change could reduce the mismatch? | Link analysis to governance |

## V. An Age-Sensitive Contextual Integrity Framework

This section proposes a simple framework for age-sensitive privacy analysis. It uses Contextual Integrity without changing the five parameters and without adding age as a sixth. Instead, it uses age to ask whether the same flow means different things to different users. The framework has four steps.

*Step 1: Identify the information flow.* Name the sender, recipient, subject, attribute, and transmission principle. One user action can start several flows: a user posts to followers, the platform stores the content, the system recommends it, and other users reshare it. The analyst should separate intended sharing from later movement.

*Step 2: Find the assumed user baseline.* Ask what user the design assumes. Many privacy tools assume someone who reads settings, understands recommendation, and knows how later data use works, which is close to a digitally skilled adult [8][15].

*Step 3: Compare age-based interpretation.* Ask whether users at different life stages read the flow differently across visibility, imagined audience, persistence, and reuse. A gap between expected and actual flow signals a possible mismatch.

*Step 4: Link mismatch to design and governance.* Connect the mismatch to concrete changes: clearer audience previews, simpler controls, age-appropriate notices, stronger defaults for younger users, warnings before wider sharing, and plainer explanations of recommendation.

Table II summarizes the steps. The framework is preliminary and is meant to guide early research rather than provide final measurement. It turns age-sensitive privacy concerns

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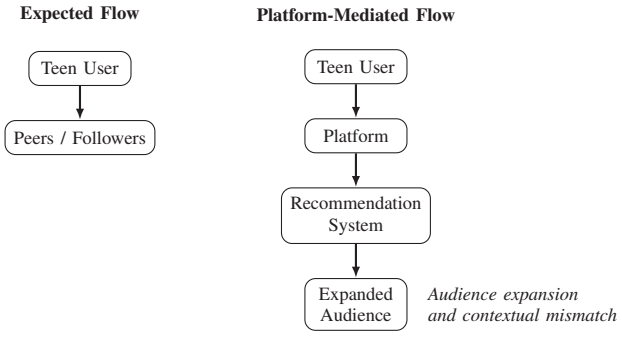



Figure 1. Expected and platform-mediated information flows in the teen visibility case.

into questions that can drive interviews, surveys, platform audits, and design reviews. Such mismatches may persist not because platforms deliberately seek to create privacy harms, but because uniform privacy designs can serve broader technical and organizational priorities. Standardized interfaces and defaults are easier to apply across large and diverse user populations, while additional warnings, audience restrictions, or age-sensitive controls can introduce friction into sharing and recommendation processes. Platform incentives may therefore favor scalable and relatively uniform privacy mechanisms even when users interpret the resulting information flows differently. From a governance perspective, normative flattening helps make this trade-off visible by asking when efficiency and uniformity come at the cost of contextual understanding.

## VI. Case Illustration: Teen Visibility and Algorithmic Amplification

We apply the framework to teen visibility and algorithmic amplification. A teen posts a short video to an audience they imagine as friends, followers, and classmates. The recommendation system may then show the video to a much wider audience, including strangers, adults, and users outside the first social circle.

From the platform's view, the flow follows the rules: the user chose to post, and the terms of service permit recommendation. From the user's view, the meaning differs, since the post felt like peer communication rather than public exposure. The mismatch appears when the system widens the audience beyond the expected social boundary.

Using the Contextual Integrity vocabulary, the initial flow can be represented as follows: the sender and subject are the teen user, the attribute is the posted video, the intended recipients are peers or followers, and the transmission principle is the user's decision to share within that perceived social boundary. The platform-mediated flow changes the recipient structure by introducing the platform and its recommendation system as intermediaries and by extending the content to a broader audience. The resulting mismatch therefore lies not only in audience size, but also in the difference between the user's understood transmission condition and the platform's algorithmic redistribution of the content.

Figure 1 illustrates how algorithmic recommendation expands the recipient boundary beyond the user's initially perceived social context, making the location of the privacy mismatch explicit. The four-step framework makes this mismatch analytically explicit. Step 1 separates the intended flow, teen to peers, from the later flow, teen to platform to algorithmically expanded audience. Step 2 names the assumed baseline: a user who knows that posting can lead to algorithmic amplification. Step 3 asks whether and how age shapes interpretation since a teen may weigh peer visibility and social judgment more than platform reach. Step 4 maps the mismatch to design and policy options such as private accounts by default for teens, audience previews, and clearer warnings before wider distribution.

Recent changes support this focus on defaults. Instagram Teen Accounts apply stronger default settings for younger users [1], and age-appropriate design codes push platforms toward safer defaults for children [2]. These measures change the design itself rather than relying on user education alone, which supports our claim that the mismatch is a design problem and not only user error. Because the framework has not yet been empirically tested, the following expectations should be understood as hypotheses rather than results. We expect perceived privacy mismatch to increase when algorithmic recommendation expands content beyond an audience that users believe they have selected. We also expect differences across age groups to be most visible in judgments about audience boundaries, content persistence, and the meaning of recommendation-mediated redistribution. Future scenario-based studies can test whether these predicted differences are systematic.

Reducing these mismatches is unlikely to depend on a single intervention. Design measures such as safer defaults, audience previews, and clearer recommendation notices can make information flows more visible, while age-appropriate privacy education can help users better understand persistence, resharing, and algorithmic amplification. These measures should be complemented by user feedback and privacy regulation so that platform practices are evaluated not only against formal consent requirements but also against how users actually understand the resulting information flows.

The case is one example. It does not show that all teens share one expectation, nor that age is the only factor that matters. It shows how age-sensitive analysis can make a hidden design assumption visible.

## VII. Conclusion and Future Work

This work in progress argues that age-sensitive analysis can reveal privacy mismatches in social media design. Uniform settings can hide different user expectations. A privacy rule may look neutral while still assuming a digitally skilled adult who understands settings, recommendation, and long-term data use.

This matters for governance. Platforms point to consent tools, account settings, and policies as proof of user control. Those tools are useful, but they may fall short when users understand flows in different ways. Audience previews, simpler controls,

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clearer explanations of recommendation, and age-appropriate defaults can help users see how content may travel beyond the audience they first imagined.

The paper has several limitations. It is conceptual and does not yet include user-study data, so the proposed mismatches should not be interpreted as established differences between age groups. Age is also only one source of variation in privacy expectations; culture, digital skill, prior experience, and platform-specific context may shape how the same information flow is understood. In addition, the framework has so far been illustrated with a limited set of social-media scenarios and has not yet been tested across platforms or regulatory contexts. These limitations motivate the empirical validation described above. Even so, the framework provides an early and testable way to identify where shared platform designs may produce different privacy experiences across users.

Future work will evaluate the framework through scenario-based studies that systematically vary audience scope, algorithmic recommendation, content persistence, and transmission principles. Participants from different age groups will evaluate each scenario in terms of expectedness, contextual appropriateness, perceived privacy risk, perceived control, and understanding of audience expansion. The analysis will compare responses across age groups and information-flow conditions to identify where the largest expectation–behavior mismatches occur and whether the proposed flattening categories consistently identify those mismatches. Interviews and platform audits will complement the scenario ratings by examining how users interpret privacy settings and how platform defaults shape privacy experience. Because this paper presents the framework as work in progress, this empirical validation remains a future phase of the study rather than a source of results reported here.

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